# List of Publications


[1] E.D. Sontag. *Notes on Mathematical Systems Biology (online).* Online only., 2026. Continuously updated. If the link does not work, then copy/paste this: http://drive.google.com/drive/folders/1lIRqaCPeXMVZGoY-44bBsvtnsHtlRfIO?usp=sharing.

[2] E. D. Sontag. Dynamics and dose response in scaffold ligand binding. 2026. Submitted. Preprint in arXiv:2508.06599.

[3] K. Manoj, D. Jatkar, M. Ali Al-Radhawi, E.D. Sontag, and D. Del Vecchio. Paradoxical gene regulation explained by competition for genomic sites. 2026. Submitted. Also bioRxiv 2025.11.27.691022.

[4] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. Boundedness of solutions in feedback systems with antithetic controllers. 2026. Submitted. Preprint in arXiv 2604.27290.

[5] M.K. Wafi, A.C.B de Oliveira, and E.D. Sontag. When is cumulative dose response monotonic? analysis of incoherent feedforward motifs. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. To appear. Also arXiv:2604.01573.

[6] A. Oliveira, A. C. B. de Oliveira, M. Sznaier, and E. D. Sontag. On incremental and semi-global exponential stability of gradient flows satisfying generalized lojasiewicz inequalities. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. To appear. Also arXiv arXiv:2603.25822.

[7] A. C. B. de Oliveira, R. Wang, I.R. Manchester, and E. D. Sontag. Remarks on Lipschitz-minimal interpolation: Generalization bounds and neural network implementation. In *Proc. 65th IEEE Conference on Decision and Control (CDC)*, 2026. To appear. Also arXiv:2603.19524.

[8] E.D. Sontag. Dynamic response phenotypes and model discrimination in systems and synthetic biology. 2025. Submitted. Preprint in arXiv 2512.24945.

[9] D.D. Jatkar, K. M. Aravind, E. D. Sontag, and D. Del Vecchio. Paradoxical gene regulation explained by competition for genomic sites. *bioRxiv, being submitted.*, page 2025.11.27.691022, 2025.

[10] A.P. Tran, D.D. Jatkar, M.A. Al-Radhawi, E. Ernst, and E.D. Sontag. Optimization of heuristic logic synthesis by iteratively reducing circuit substructures using a database of optimal implementations. *bioRxiv*, page 2025.11.28.691216, 2025.

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